



Cloud Computing

Assignment- Week 0

TYPE OF QUESTION: MCQ/MSQ

Number of questions: 10

Total mark: 10 X 1 = 10

QUESTION 1:

Which of the following port numbers fall in the ephemeral/private (dynamic) range?

- a) 80
- b) 1023
- c) 49259
- d) 55000

Correct Answer: (c), (d)

Detailed Solution: Ephemeral (dynamic/private) ports are in the range 49152–65535. Port 49152 and 55000 lie in that range. Port 80 is well-known (HTTP) and 1023 is still in the well-known system range (0–1023).

QUESTION 2:

Which of the following devices operates at the Network Layer (Layer 3)?

- a) Hub
- b) Gateway
- c) Repeater
- d) Switch

Correct Answer: (b)

Detailed Solution: A gateway operates at the Network Layer and can connect networks with different protocols. A hub and repeater operate at the Physical Layer (Layer 1), while a switch operates at the Data Link Layer (Layer 2).

QUESTION 3:

In classful addressing, the IP address 172.16.5.4 belongs to which class??

- a) Class A
- b) Class B
- c) Class C
- d) Class D

Correct Answer: (b)

Detailed Solution: Classful ranges: A = 1.0.0.0–126.255.255.255, B = 128.0.0.0–191.255.255.255, C = 192.0.0.0–223.255.255.255. IP 172.16.5.4 is within 128–191 hence Class B

QUESTION 4:

In the OSI model, which layer is primarily responsible for data representation, encryption, and translation between application formats?

- a) Session Layer
- b) Presentation Layer
- c) Transport Layer
- d) Application Layer

Correct Answer: (b)

Detailed Solution: The Presentation layer (Layer 6) handles syntax translation, data representation (e.g., character encoding), and optional encryption/compression i.e., preparing data for the Application layer. Session manages dialog control; Transport handles end-to-end delivery.

QUESTION 5:

Which of the following application-layer protocols uses TCP as its transport layer protocol for reliable file transfer?

- a) SNMP
- b) TFTP
- c) FTP
- d) DHCP

Correct Answer: (c) FTP

Detailed Solution: FTP (File Transfer Protocol) uses TCP to ensure reliable and error-free file transmission. SNMP, TFTP, and DHCP primarily use UDP, which does not guarantee delivery.

QUESTION 6:

Match the following protocols with their respective TCP/IP layers:

- | | |
|-------------|----------------------|
| 1. ICMP | A. Data Link Layer |
| 2. TELNET | B. Application Layer |
| 3. Ethernet | C. Transport Layer |
| 4. DHCP | D. Network Layer |

- a) 1-D, 2-B, 3-A, 4-B
- b) 1-C, 2-A, 3-B, 4-D
- c) 1-B, 2-C, 3-D, 4-A
- d) 1-D, 2-C, 3-A, 4-D

Correct Answer: (a)

Detailed Solution: ICMP (Internet Control Message Protocol) is a network layer protocol (Layer 3) - Answer: D .TELNET is an application layer protocol (Layer 7) - Answer: B Ethernet is a data link layer protocol (Layer 2) - Answer: A DHCP (Dynamic Host Configuration Protocol) is an application layer protocol (Layer 7) - Answer: B

Therefore, the correct matching is 1-D, 2-B, 3-A, 4-B.

QUESTION 7:

In the OSI model, the layer responsible for logical topology and determining the best path for data transmission is:

- a) Data Link Layer
- b) Physical Layer
- c) Transport Layer
- d) Network Layer

Correct Answer: (d)

Detailed Solution: The Network Layer (Layer 3) is responsible for logical addressing (IP addresses), routing, and determining the best path for data transmission between networks using routing algorithms. The Data Link Layer (Layer 2) handles physical addressing and operates on the local network segment. The Physical Layer (Layer 1) deals with physical transmission media. The Transport Layer (Layer 4) manages end-to-end communication reliability.

QUESTION 8:

The transport layer protocols commonly used for the following services — (real-time multimedia, email, file transfer, DNS) respectively are:

- a) TCP, UDP, UDP, TCP
- b) UDP, TCP, TCP, UDP
- c) TCP, TCP, UDP, TCP
- d) UDP, UDP, TCP, UDP

Correct Answer: (b) UDP, TCP, TCP, UDP

Detailed Solution: Real-time multimedia typically uses UDP (lower latency, tolerance for loss). Email (SMTP/POP/IMAP) uses TCP. File transfer (FTP/SFTP) uses TCP. DNS typically uses UDP for queries (TCP is used for zone transfers and when responses exceed UDP limits). So the order is UDP, TCP, TCP, UDP.

QUESTION 9:

In a direct mapped cache Tag is 4bits, Index is 12 bits, block size is 1 byte. What is the size of Main Memory and Cache Memory?

- a) 64Kb and 4Kb
- b) 128Kb and 16Kb
- c) 64Kb and 8Kb
- d) 128Kb and 6Kb

Corrected Answer : (a)

Detailed Solution : Total Address Bits = Tag + Index + Block Offset
Block Offset = $\log_2(\text{Block Size}) = \log_2(1) = 0$ bits



Total Address Bits = $4 + 12 + 0 = 16$ bits

Number of Cache Lines = $2^{(\text{Index bits})}$

$$= 2^{12}$$

$$= 4,096 \text{ cache lines}$$

Cache Size = Number of Cache Lines $\times$ Block Size

$$= 4,096 \times 1 \text{ Byte}$$

$$= 4,096 \text{ Bytes}$$

$$= 4 \text{ KB}$$

Main Memory Size = $2^{(\text{Total Address Bits})}$

$$= 2^{16}$$

$$= 65,536 \text{ Bytes}$$

$$= 64 \text{ KB}$$

QUESTION 10:

In a 8-way Set Associative Mapping, Cache Memory is 64Kb, Main Memory Address is 32bit. The address is sub divided into Tag, Index and Block Offset. The number of bits in the Tag is :

a)20

b)19

c)18

d)22

Correct Answer: (b) 19

Detailed Solution : Block Size = 64 Bytes = 2^6 Bytes

Block Offset = $\log_2(64) = 6$ bits .Cache Size = Number of Sets $\times$ Associativity $\times$ Block Size

$$64 \text{ KB} = \text{Number of Sets} \times 8 \times 64 \text{ Bytes. Number of Sets} = 64 \text{ KB} / (8 \times 64 \text{ Bytes}) = 65,536 / 512$$

$$= 128 \text{ sets} = 2^7 \text{ sets .Index Bits} = \log_2(128) = 7 \text{ bits .Total Address Bits} = \text{TAG} + \text{INDEX} + \text{BLOCK OFFSET. Therefore, TAG} = 32 - 7 - 6 = 19 \text{ bits}$$
